

SECTION 5.2

1. Label the following statements as true or false.

(a) Any linear operator on an n -dimensional vector space that has fewer than n distinct eigenvalues is not diagonalizable.

Answer : False. If $T = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, then T has only one eigenvalue 1 but T is diagonalizable.

(b) Two distinct eigenvectors corresponding to the same eigenvalue are always linearly dependent.

Answer : False. If $T = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $v_1 = (1, 0)$ and $v_2 = (0, 1)$, then these two eigenvectors v_1, v_2 corresponding to eigenvalue 1 is linearly independent.

(c) If λ is an eigenvalue of a linear operator T , then each vector in E_λ is an eigenvector of T .

Answer : False. The zero vector is not eigenvector by definition.

(d) If λ_1 and λ_2 are distinct eigenvalues of a linear operator T , then $E_{\lambda_1} \cap E_{\lambda_2} = \{0\}$.

Answer : True. Let v be a nonzero vector such that $v \in E_{\lambda_1} \cap E_{\lambda_2}$. Then $\lambda_1 v = \lambda_2 v$. This implies that $(\lambda_1 - \lambda_2)v = 0$. Since λ_1 and λ_2 are different numbers, it follows that $\frac{1}{(\lambda_1 - \lambda_2)}(\lambda_1 - \lambda_2)v = v = 0$. This contradiction shows that $E_{\lambda_1} \cap E_{\lambda_2} = \{0\}$.

(e) Let $A \in M_{n \times n}(F)$ and $\beta = \{v_1, v_2, \dots, v_n\}$ be an ordered basis for F^n consisting of eigenvectors of A . If Q is the $n \times n$ matrix whose j th column is v_j ($1 \leq j \leq n$), then $Q^{-1}AQ$ is a diagonal matrix.

Answer : True.

(f) A linear operator T on a finite-dimensional vector space is diagonalizable if and only if the multiplicity of each eigenvalue λ equals the dimension of E_λ .

Answer : False. See Theorem 5.11.

(g) Every diagonalizable linear operator on a nonzero vector space has at least one eigenvalue.

Answer : True. See Theorem 5.11.

(h) If a vector space is the direct sum of subspaces $W_1, W_2, \dots, W_k$, then $W_i \cap W_j = \{0\}$ for $i \neq j$.

Answer : True.

2. For each of the following matrices $A \in M_{n \times n}(R)$, test A for diagonalizability, and if A is diagonalizable, find an invertible matrix Q and a diagonal matrix D such that $Q^{-1}AQ = D$.

(a) $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$

It is not diagonalizable since it has only one 1-dimensional eigenspace E_{λ_1} corresponding to only one eigenvalue $\lambda_1 = 1$.

(d) $\begin{pmatrix} 7 & -4 & 0 \\ 8 & -5 & 0 \\ 6 & -6 & 3 \end{pmatrix}$

Note that its characteristic polynomial is $(3 - x)(x - 3)(x + 1)$. Thus, its eigenvalues are $\lambda_1 = 3$ and $\lambda_2 = -1$. Then, we can find eigenvectors $(1, 1, 0)$, $(1, 1, 1)$ for E_{λ_1} and

$(2, 4, 3)$ for E_{λ_2} . We can see that $D = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -1 \end{pmatrix}$ and $Q = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 1 & 4 \\ 0 & 1 & 3 \end{pmatrix}$. (D and Q

are not the unique solution here.)

(e) $\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & -1 \\ 0 & 1 & 1 \end{pmatrix}$

It has characteristic polynomial $(x - 1)(-x^2 - 1)$. Since it does not split, the matrix is not diagonalizable.

3. For each of the following linear operators T on a vector space V , test T for diagonalizability, and if T is diagonalizable, find a basis β for V such that $[T]_\beta$ is a diagonal matrix.

(a) $V = P_3(R)$ and T is defined by $T(f(x)) = f'(x) + f''(x)$, respectively.

(d) $V = P_2(R)$ and T is defined by $T(f(x)) = f(0) + f(1)(x + x^2)$.

(e) $V = C^2$ and T is defined by $T(z, w) = (z + iw, iz + w)$.

(f) $V = M_{2 \times 2}(R)$ and T is defined by $T(A) = A^t$.

Proof : Note that the answer to this question may vary depending on the basis and its order.

(a) Let β be the ordered standard basis $\{1, x, x^2, x^3\}$ for $V = P_3(R)$. Then $[T]_\beta$ is
$$\begin{pmatrix} 0 & 1 & 2 & 0 \\ 0 & 0 & 2 & 6 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$
. The characteristic polynomial of this matrix is x^4 . Hence, T has only one eigenvalue 0. It is easy to see that the basis of E_0 is $\{(1, 0, 0, 0)\}$. Since $\dim E_0 = 1 \neq 4$, T is not diagonalizable.

(d) Let β be the ordered standard basis $\{1, x, x^2\}$ for $V = P_2(R)$. Then $[T]_\beta$ is
$$\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}$$
. The characteristic polynomial of this matrix is $(1 - x)(x)(x - 2)$. Thus, T has eigenvalues 0, 1, 2. Note that $(0, 1, -1)$, $(1, 0, -1)$, $(0, 1, 1)$ are basis of E_0, E_1, E_2 , respectively. Then $D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ and $Q = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ -1 & -1 & 1 \end{pmatrix}$.

(e) Let β be the ordered standard basis $\{(1, 0), (0, 1)\}$ for $V = C^2$. Then $[T]_\beta$ is
$$\begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}$$
. The characteristic polynomial of this matrix is $x^2 - 2x + 2$. Thus, T has eigenvalues $1 - i, 1 + i$. Note that $(1, -1), (1, 1)$ are basis of E_{1-i}, E_{1+i} , respectively. Then $D = \begin{pmatrix} 1 - i & 0 \\ 0 & 1 + i \end{pmatrix}$ and $Q = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$.

(f) Let β be the ordered standard basis $\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$ for $V = M_{2 \times 2}(R)$. Then $[T]_\beta$ is
$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$
. The characteristic polynomial of this matrix

is $(1 - x)^2(x^2 - 1)$. Thus, T has eigenvalues 1 and -1 . Note that $\left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$

are basis of E_1 . Also, $\left\{ \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right\}$ is a basis of E_{-1} . Then $D = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix}$

and $Q = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$.

7. For

$$A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} \in M_{2 \times 2}(R),$$

find an expression for A^n , where n is an arbitrary positive integer.

Proof : First, the characteristic polynomial of A is $(1-x)(3-x) - 8 = x^2 - 4x - 5$. Thus, A has two eigenvalues, $\lambda_1 = 5$ and $\lambda_2 = -1$. Choose $v_1 = (1, 1)$ and $v_2 = (-2, 1)$. Then they are bases for the eigenspaces E_{λ_1} and E_{λ_2} . Let $Q = \begin{pmatrix} 1 & -2 \\ 1 & 1 \end{pmatrix}$ (its i -th column is corresponding to v_i). Then, we have $Q^{-1}AQ = \begin{pmatrix} 5 & 0 \\ 0 & -1 \end{pmatrix}$. Therefore,

$$A^n = Q \begin{pmatrix} 5^n & 0 \\ 0 & (-1)^n \end{pmatrix} Q^{-1}.$$

8. Suppose that $A \in M_{n \times n}(F)$ has two distinct eigenvalues, λ_1 and λ_2 , and that $\dim(E_{\lambda_1}) = n - 1$. Prove that A is diagonalizable.

Proof : Note that $\dim(E_{\lambda_2}) \geq 1$. Let V be the n -dimensional vectorspace where A is a linear operator on V . Since $\dim(E_{\lambda_1}) = n - 1$, we have $V = E_{\lambda_1} \oplus E_{\lambda_2}$. Thus, A is diagonalizable.

9. Let T be a linear operator on a finite-dimensional vector space V , and suppose there exists an ordered basis β for V such that $[T]_\beta$ is an upper triangular matrix.

(a) Prove that the characteristic polynomial for T splits.

Proof : Recall that the characteristic polynomial of T is independent of the choice of basis β . Indeed,

$$\det([T]_\beta - xI) = \det(Q^{-1})\det([T]_\beta - xI)\det(Q) = \det(Q^{-1}[T]_\beta Q - xI)$$

where I is the identity matrix and Q is an invertible matrix. Thus, the characteristic polynomial for T splits as $\prod_{i=1}^n ([T]_{\beta ii} - x)$ where $[T]_{\beta ii}$ denotes ii -entry of the matrix $[T]_{\beta}$.

10. Let T be a linear operator on a finite-dimensional vector space V with the distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_k$ and corresponding multiplicities $m_1, m_2, \dots, m_k$. Suppose that β is a basis for V such that $[T]_{\beta}$ is an upper triangular matrix. Prove that the diagonal entries of $[T]_{\beta}$ are $\lambda_1, \lambda_2, \dots, \lambda_k$ and that each λ_i occurs m_i times ($1 \leq i \leq k$).

Proof : Consider the characteristic polynomial $f(t)$ of $[T]_{\beta}$. The roots of $f(t)$ is diagonal entries of $[T]_{\beta}$ since $[T]_{\beta}$ is an upper triangular matrix. Also, the roots of $f(t)$ correspond to eigenvalues of T . This implies that the diagonal entries of $[T]_{\beta}$ are $\lambda_1, \lambda_2, \dots, \lambda_k$ and that each λ_i occurs m_i times ($1 \leq i \leq k$).

11. Let A be an $n \times n$ matrix that is similar to an upper triangular matrix and has the distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_k$ with corresponding multiplicities $m_1, m_2, \dots, m_k$. Prove the following statements.

(a) $\text{tr}(A) = \sum_{i=1}^k m_i \lambda_i$

(b) $\det(A) = (\lambda_1)^{m_1} (\lambda_2)^{m_2} \dots (\lambda_k)^{m_k}$.

Proof : These results follow immediately from the fact that trace of A is equal to the coefficient of t^{n-1} of the characteristic polynomial $f(x)$ of A , and $\det(A)$ is equal to the constant term of the characteristic polynomial $f(x)$ of A .

12. Let T be an invertible linear operator on a finite-dimensional vector space V .
 (a) Recall that for any eigenvalue λ of T , λ^{-1} is an eigenvalue of T^{-1} (Exercise 8 of Section 5.1). Prove that the eigenspace of T corresponding to λ is the same as the eigenspace of T^{-1} corresponding to λ^{-1} . (b) Prove that if T is diagonalizable, then T^{-1} is diagonalizable.

Proof : (a) Let $v \in V$. If $T(v) = \lambda v$, then $v = T^{-1}(\lambda v)$. This can be rewritten as $T^{-1}(v) = \frac{1}{\lambda} v$. Similarly, it is easy to see that if $T^{-1}(v) = \lambda^{-1} v$, then $T(v) = \lambda v$. This completes the proof for (a).

(b) Recall that T is diagonalizable if and only if V is the direct sum of the eigenspaces of T . Since the eigenspace of T corresponding to the eigenvalue λ of T is the same as eigenspace of T^{-1} corresponding to the eigenvalue λ^{-1} of T^{-1} , we can say that if T is diagonalizable, then T^{-1} is diagonalizable.

18. (a) Prove that if T and U are simultaneously diagonalizable operators, then T and U commute (i.e., $TU = UT$). (b) Show that if A and B are simultaneously diagonalizable matrices, then A and B commute.

Proof : (a) Since T and U are simultaneously diagonalizable operators, there exists an ordered basis β for V such that $[T]_\beta$ and $[U]_\beta$ are diagonal matrices. Since diagonal matrices commute, T and U commute.

(b) There exists an invertible matrix Q such that $Q^{-1}AQ$ and $Q^{-1}BQ$ are diagonal matrices. Since diagonal matrices commute, we have $Q^{-1}AQQ^{-1}BQ = Q^{-1}BQQ^{-1}AQ$. This implies that $Q^{-1}ABQ = Q^{-1}BAQ$. Since Q is invertible, we have $AB = BA$.

19. Let T be a diagonalizable linear operator on a finite-dimensional vector space, and let m be any positive integer. Prove that T and T^m are simultaneously diagonalizable.

Proof : Let T be a linear operator on an n -dimensional vector space V . Since T is diagonalizable, there exists an ordered basis β for V such that $[T]_\beta$ is a diagonal matrix. Note that $[T]_\beta^m = [T^m]_\beta$. Since n th power of a diagonal matrix is still a diagonal matrix, we conclude that T and T^m are simultaneously diagonalizable.